

posterior tubercles of transverse processes of **C2–C6**. It is inserted into the **first rib**, posterior to the insertion of the anterior scalene muscle.

- The **posterior scalene muscle** has the same origins as the middle scalene muscle, but it inserts on the **second rib**.

Lecture (3): Nerves of the head & neck I

Olfactory nerve (CN I):

- The **olfactory nerve** is a special somatic afferent nerve which innervates the olfactory mucosa within the nasal cavity. It carries information about smell to the brain.
- It is formed of bundles of nerve fibers of the **olfactory receptor cell axons, which** pass through the cribriform plate of the ethmoid bone. It terminates in the olfactory bulb in the cranial cavity.

Optic nerve (CN II):

- **Optic nerve** is a special somatic afferent nerve which innervates the retina of the eye and brings visual information to the brain.
- Each **optic nerve** is formed of axons of **retinal ganglion cells**.
- Its course can be subdivided into **four parts**:
 - **Intraocular part** is the most anterior part that is located within the eyeball
 - **Intraorbital part** from the posterior part of the eyeball to the optic canal.
 - **Intracanalicular part**: within the optic canal.
 - **intracranial part**: from optic canal to optic chiasma.
- It travels through **optic canal** to form the **optic chiasma**.
- The **nasal retinal fibers** of each nerve cross to the opposite side forming the **optic chiasma** and enter the **contralateral optic tract**, whereas those of each **temporal retinal fiber** remain in the same side and enter **ipsilateral optic tract**.
- **Optic tract fibers** relay in **lateral geniculate body (LGB)**, axons of **LGB** form optic radiation which terminate on **primary visual cortex** in the lips of **calcarine sulcus**.

Oculomotor nerve (CN III):

- It consists of both a **somatic and visceral efferent motor nerve**.
- It originates from the **motor nucleus & Edinger Westphal nucleus (parasympathetic)** in the upper part of midbrain.
- It leaves the **cranial cavity** through the **superior orbital fissure** to enter the **orbit** by dividing into **superior & inferior divisions**.
- It supplies **all extra-ocular muscles** except the superior oblique & lateral rectus. Also to **intra-ocular muscles**: constrictor pupillary muscle & ciliary muscle via parasympathetic (constriction of pupil and lens accommodation).
- **Lesion: lateral squint, pupil dilatation &**

ptosis. Trochlear nerve (CN IV):

- The **trochlear nerve** is a **general somatic motor nerve**.
- It originates from the motor nucleus in the lower part of midbrain.
- It enters the **orbit** through the superior orbital fissure, supplying **superior oblique muscle**.
- **Lesion: diplopia**

Trigeminal nerve (CN V)

- **Trigeminal nerve** is a mixed (motor and sensory).
- It is attached to pons by two roots: **larger sensory root** and **smaller motor one**.
- It has **one motor nucleus** and **3 sensory nuclei** (Main sensory nucleus of trigeminal nerve, Spinal nucleus of trigeminal nerve & Mesencephalic nucleus of trigeminal nerve).
- It joins the trigeminal ganglion at the apex of the petrous bone.
- The trigeminal ganglion gives rise to **3 branches**:

Ophthalmic nerve V1:

- Pure sensory nerve of the orbit. From the trigeminal ganglion.
- **It gives 3 branches** before passing through superior orbital fissure into orbit:

1- **Frontal nerve:** It enters the orbit through the superior orbital fissure.

- The frontal nerve divides into **supratrochlear** and **supraorbital** branches supplying skin of forehead.

2- **Lacrimal nerve:** It enters the orbit through the superior orbital fissure.

- It is sensory to lacrimal gland.

3- **Naso ciliary nerve:** It enters the orbit through the superior orbital fissure. It gives of:

- **Sensory root** to the ciliary ganglion.
- **Long ciliary nerves** are sensory to ciliary body, iris & cornea.
- **Posterior ethmoidal nerve** supplies mucosa of ethmoidal air & sphenoid sinuses.
- **Infratrochlear nerve** supplies medial parts of eyelids and upper part of the nose.
- **Anterior ethmoidal nerve** supplies mucosa of ethmoid air sinus and lower part of the lateral wall of the nose.

Maxillary nerve V2:

➤ Pure sensory nerve, From the trigeminal ganglion.

➤ **Course:**

○ Leaves the cranial cavity through the **foramen rotundum** to the **pterygopalatine fossa**.

○ It enters the orbit through the inferior orbital fissure as infraorbital nerve which lies in the infra-orbital groove and canal. It appears in the face through infraorbital foramen.

○ **Branches** From the **trunk**:

- **Meningeal branch** supplies dura of middle cranial fossa.
- **Ganglionic branches** that suspend the **pterygopalatine ganglion** (parasympathetic ganglion which supplies the lacrimal gland & nasal glands by Secreto-motor fibers).
- **Posterior superior alveolar (PSA):** supplies molar teeth.

- **Zygomatic nerve:** gives off zygomatico-facial & zygomatico-temporal supplying skin of the face & scalp.

o Branches from Infra-orbital nerve:

- **Middle superior alveolar nerve** supplies the premolars.
- **Anterior superior alveolar nerve** supplies the canines and incisors.
- **Terminal branches:**

- 1- **Labial:** to upper lip.
- 2- **Nasal:** to skin of the side of the nose.
- 3- **Palpebral:** to skin of the lower eye lid.

Mandibular nerve V3:

▣ It is a **mixed** nerve (motor and sensory fibers).

▣ **Course:** From the trigeminal ganglion & leaves the cranial

cavity through the **foramen ovale**. It ends by dividing into two divisions (anterior & posterior) in infratemporal fossa.

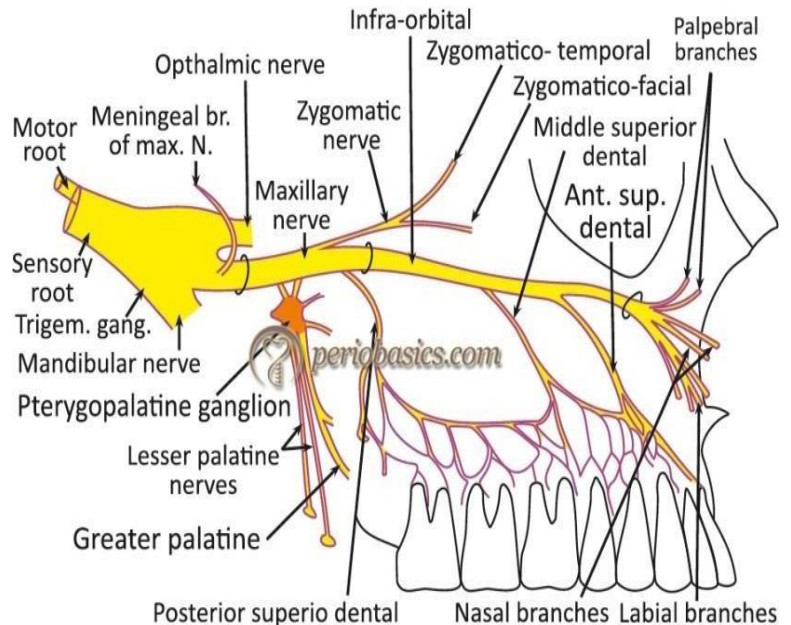
▣ **Branches:**

- From the **trunk:**

- 1- **Nervous spinosus:** through the foramen spinosum, back to cranial cavity to supply dura mater of middle cranial fossa.
- 2- **Nerve to medial pterygoid:** it supplies medial pterygoid muscle & tensor palati & tensor tympani.

- **Branches of the anterior division (3 motor, 1 sensory):**

- 1- Nerve to masseter.
- 2- Two deep temporal nerves to temporalis.



3- Nerve to lateral pterygoid.

4- Buccal branch sensory innervates the skin of the cheek and buccal mucosa.

- **Branches of the posterior division (3 sensory, 1 motor):**

1- **Auriculo-temporal nerve (sensory):** it originates by **two roots** around the middle meningeal artery. It passes through the parotid gland. It supplies auricle, parotid, temple skin & TMJ.

2- **Lingual nerve:** it unites with the chorda tympani. Then the nerve lies in direct contact with the medial aspect of the **socket** of the **lower 3rd molar tooth**. The submandibular parasympathetic ganglion is suspended to the **lingual nerve**. It is sensory to the **anterior 2/3** of the **tongue**, **floor of mouth** and carries **parasympathetic supply** to submandibular and sublingual salivary glands.

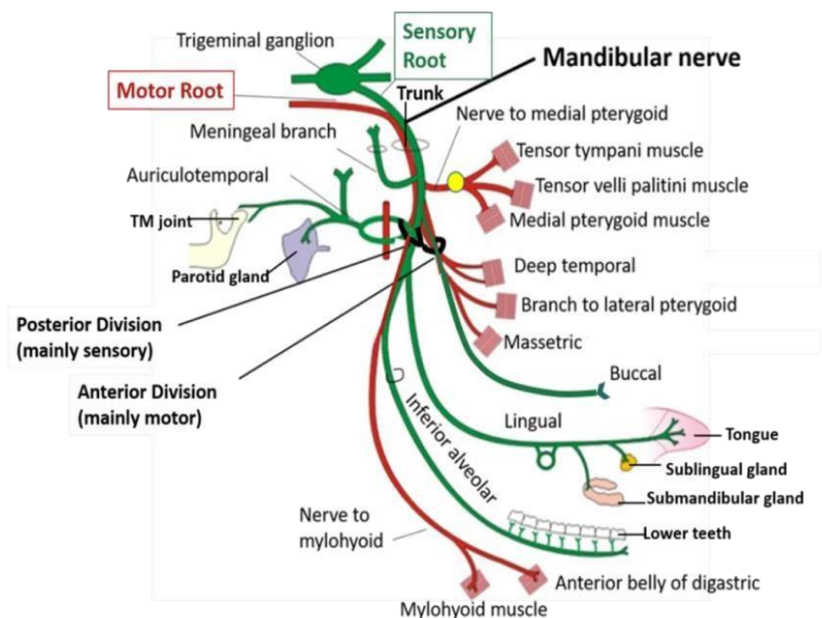
3- **Inferior alveolar**

nerve: it enters the mandibular canal. It is concerned with the sensory supply of the **teeth of the lower jaw**. It gives off

mental nerve

that appears at mental foramen and supplies skin of the **chin and lower lip**.

4- **Nerve to mylohyoid** from the inferior alveolar to supply the **mylohyoid muscle & anterior belly of digastric**.



Abducent nerve (CN VI):

- The **abducent nerve** is a **general somatic efferent nerve**.

- It originates from the motor nucleus of the **lower pons**.
- It enters the **orbit** through the superior orbital fissure.
- It innervates the **lateral rectus muscle**.
- Lesion: **medial squint**.

Facial nerve (CN VII)

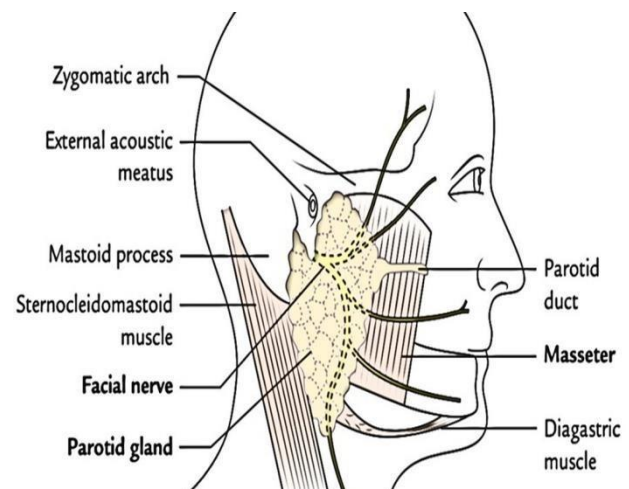
- Facial nerve is a **mixed nerve**.
- It has a motor nucleus, solitary nucleus (sensory) & superior salivatory nucleus (parasympathetic).
- It originates from the lower pons by two roots: a larger **motor root** and a smaller root (**nervus intermedius**) carrying sensory and parasympathetic fibers.
- The two roots leave the cranial cavity through the internal acoustic meatus.
- The roots travel through facial canal where they join forming the **facial nerve** properly.
- It leaves the cranium through the **stylomastoid foramen**.
- Facial nerve enters the substance of parotid gland.
- It gives **5 terminal branches** which emerge from the anterior border of parotid gland.
- **Branches:**
 - **in the facial canal:**

1- Greater petrosal nerve carries parasympathetic fibers to the pterygopalatine ganglion. Postganglionic branches supply **lacrimal, nasal, palatine & pharyngeal glands**.

2- Nerve to stapedius muscle.

3- Chorda tympani: it joins the lingual nerve.

It carries **parasympathetic fibers** to submandibular ganglion. Postganglionic fibers



supply the submandibular gland & also **taste fibers** to the anterior 2/3 of the tongue.

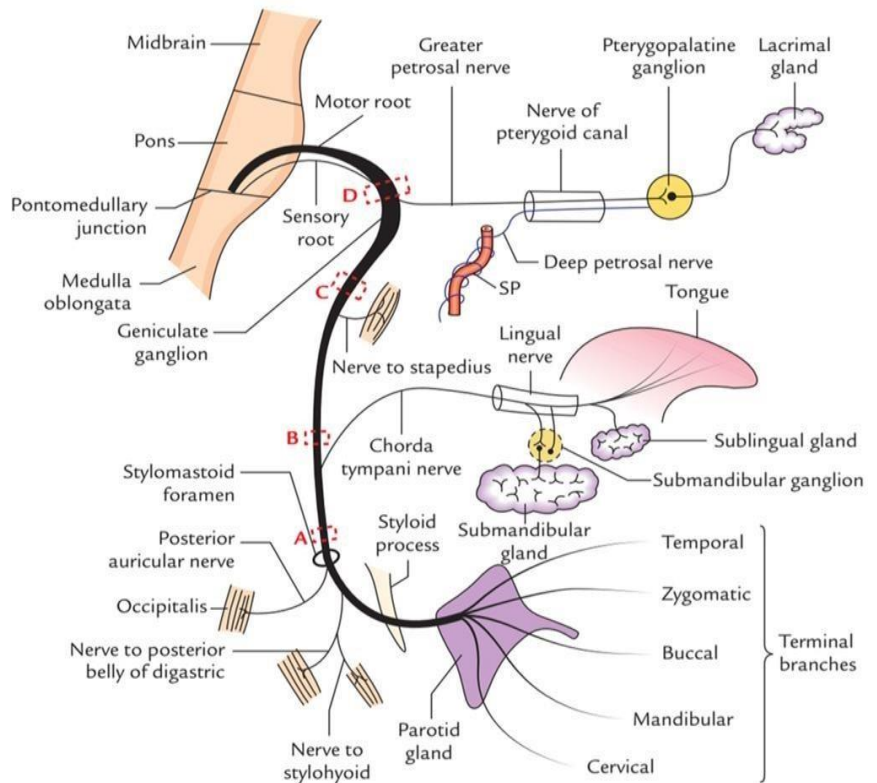
○ Extracranial part:

- Just below stylomastoid foramen, it gives off:

- 1- **posterior auricular nerve** supplies occipitalis muscle.
- 2- **Nerve to posterior belly of digastric and stylohyoid muscle.**

- It gives **five terminal** branches within the parotid gland:

- 1- **Temporal nerve** supplies frontalis and orbicularis oculi.
- 2- **Zygomatic nerve** supplies orbicularis oculi muscle.
- 3- **Buccal nerve** supplies buccinator & muscles of upper lip.
- 4- **Marginal mandibular nerve** supplies muscles of lower lip.
- 5- **Cervical nerve** supplies platysma.



Lecture (4): Nerves of the

head & neck II

Vestibulocochlear nerve (CN VIII)

- **Cranial nerve 8** is a special somatic afferent nerve.
- It is comprised of **two parts: vestibular nerve and cochlear nerve.**